Plot-a-thon

Theme : Grocery

Team: A _Square

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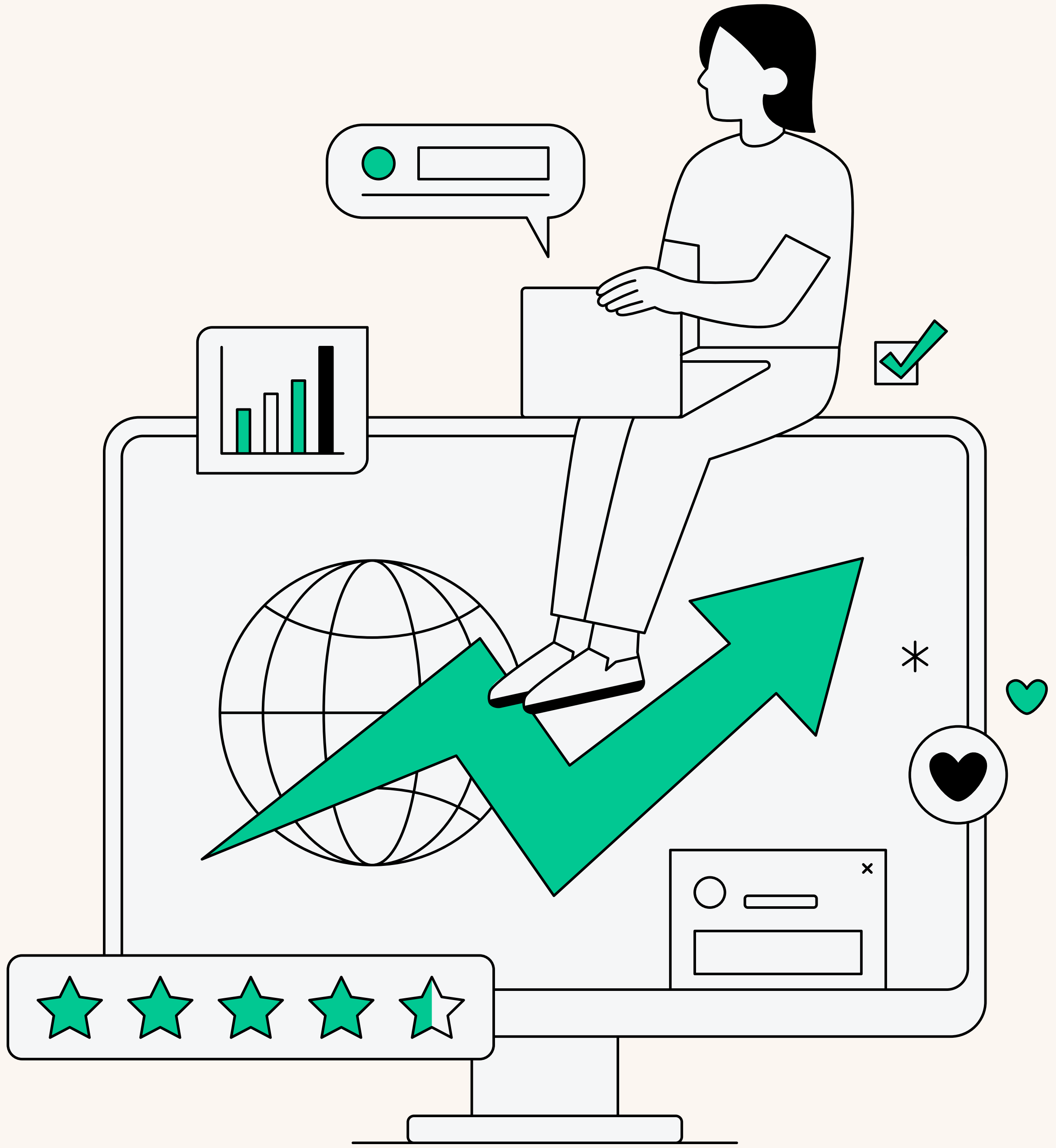
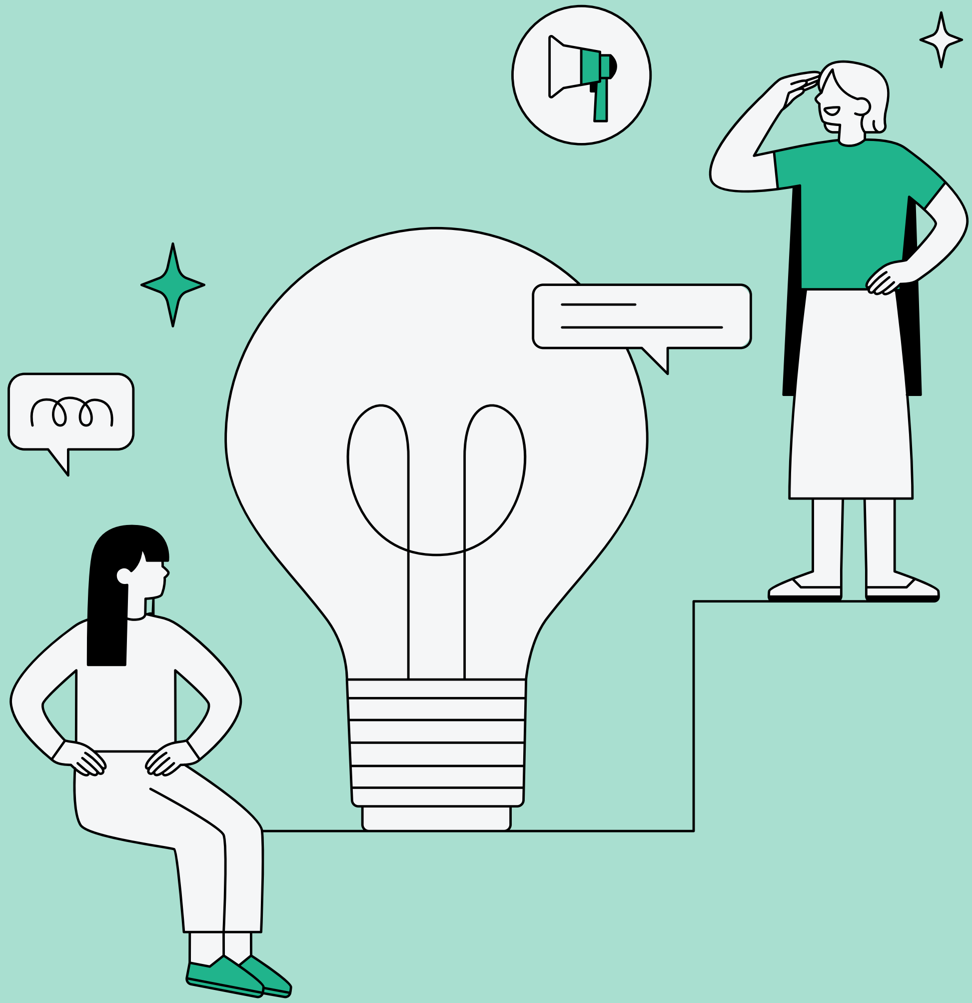


Introduction & Problem Statement

The Plot-A-Thon project involves merging and analyzing customer, sales, product, store, and exchange rate data to uncover valuable business insights. The goal is to preprocess the data, calculate key metrics, and create visualizations to address specific business questions. This will help in understanding customer demographics, product performance, delivery efficiency, and store sales effectiveness.



Problem Statement:

We are tasked with:

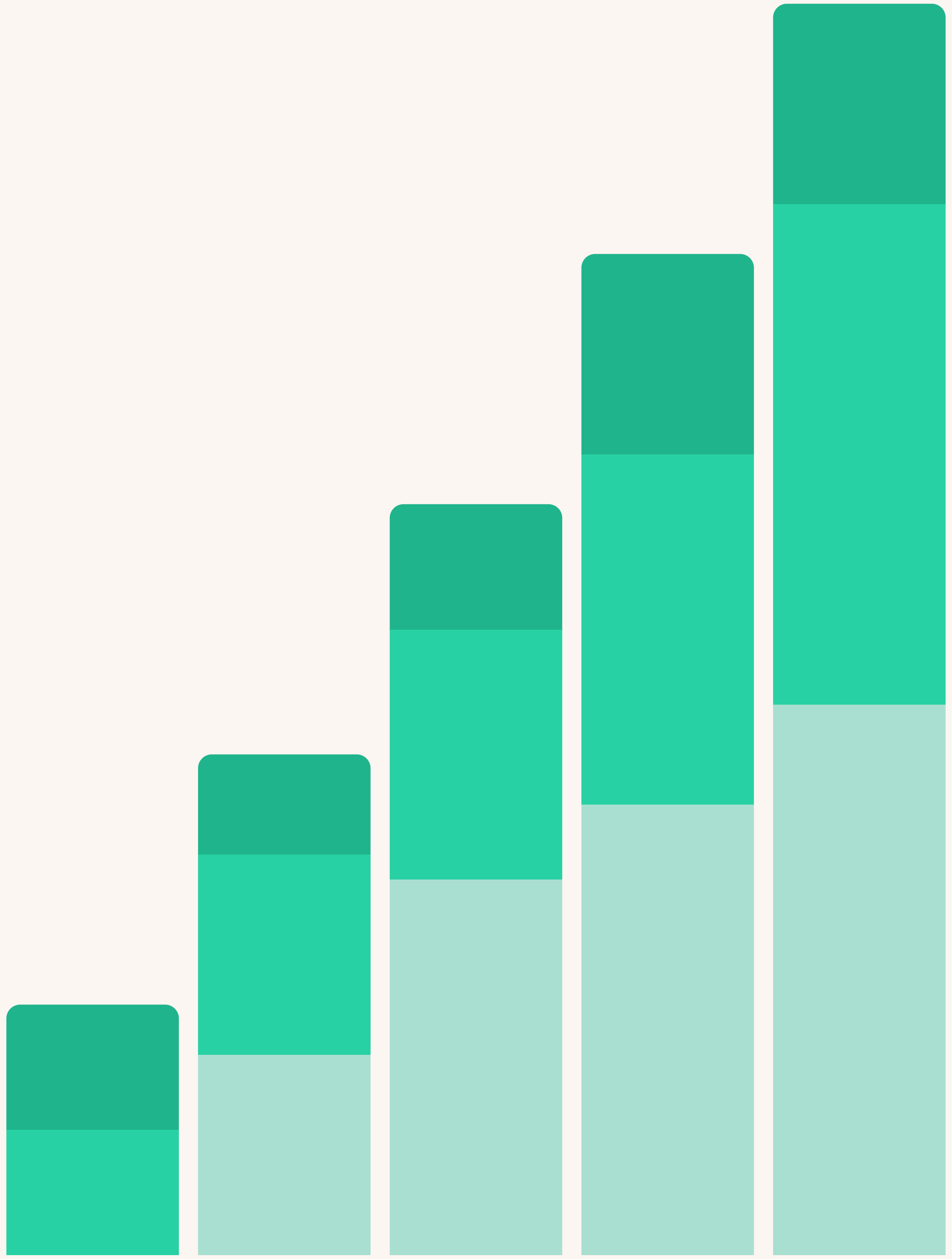
1. **Merging Datasets: Combine customer, sales, product, store, and exchange rate data using appropriate keys.**
2. **Preprocessing: Handle missing values, standardize column formats, and ensure data consistency.**
3. **Derived Metrics: Calculate total revenue, delivery time, and customer age.**
4. **Visualizations:**
   - **Customer Gender Distribution: Pie chart showing male vs. female customers.**
   - **Product Category Revenue: Tree map displaying revenue share by product category.**
   - **Delivery Time Analysis: Box plot of delivery times by state.**
   - **Top 10 Products by Revenue: Bar chart of top products by revenue.**
   - **Store Performance: Scatter plot showing store size vs. total revenue and order volume.**

The project will provide insights into sales performance, customer behavior, and store efficiencies.

Data Cleaning Process

1. **Sales File:**
    - The date format in the Delivery Date column has been standardized to a consistent format (YYYY-MM-DD) using a **delimiter.**
2. **Stores File:**
    - The Open Date field has been cleaned and formatted to ensure consistency in date representation.
3. **Product File:**
    - The data has been cleaned by handling missing or inconsistent values, ensuring it is ready for analysis.
4. **Exchange File:**
    - The date format in the Exchange Rate Date column has been standardized for consistency across datasets.

After cleaning, we merged the datasets and perform the necessary calculations (e.g., total revenue, delivery time, customer age) to generate the required visualizations.



Key Metrics

Total Revenue (USD):
Formula: Quantity × Unit Price in USD
Evaluates total sales value in USD.

Delivery Time:
Formula: Delivery Date - Order Date (days)
Measures the time taken for delivery.

Customer Age:
Formula: Current Date - Birthday
Calculates customer age for demographic analysis.

Product Category Revenue:
Formula: SUM(Quantity × Unit Price in USD)
Identifies revenue share by product category.

Number of Orders:
Formula: COUNT(Order ID)
Tracks total order volume.

Store Performance (Revenue per Square Meter):
Formula: Total Revenue / Square Meters
Assesses store efficiency relative to size.

Visualizations

Customer Gender Distribution

Suggestion (to increase the female customers)

- Targeted Marketing: Create campaigns tailored to female preferences and promote products relevant to them.
- Gender-Specific Promotions: Offer discounts or rewards on products popular with female customers.
- Enhanced Experience: Personalize the shopping experience and gather feedback to improve engagement.
- Women-Centric Events: Partner with female influencers and host events to increase brand connection.

Observation

- Male Customers: 50.75% of the total customer base is male.
- Female Customers: 49.25% of the total customer base is female.

This indicates a nearly balanced gender distribution, with a slight majority of male customers. The chart provides a clear view of the gender representation within the customer database

Visualizations

Product Category Revenue

Suggestion

- Promotional Discounts: Offer time-limited discounts or bundle deals to make games and toys more attractive to customers.
- Targeted Advertising: Focus marketing efforts on families, children, and hobbyists through social media ads and influencers.
- Seasonal Campaigns: Run special promotions during holidays or events (e.g., Christmas, Back-to-School) when demand for toys and games is typically higher.

Observation

- Computers have the highest revenue, indicating strong demand and higher sales in this category.
- Games and Toys have the lowest revenue, suggesting less interest or lower sales volume in these products.

Visualizations

Top 10 Products by Revenue

Suggestion

1. Reevaluate Product Appeal: Assess if the product features or positioning need improvement. Adjust product offerings to better meet market needs.
2. Targeted Marketing: Tailor marketing campaigns to highlight the unique selling points of Adventure Works, focusing on specific customer segments who might value these products.
3. Bundling: Offer Adventure Works products in bundles with other high-performing products like Desktop PCs to increase visibility and attractiveness.
4. Customer Feedback: Collect feedback from current customers to understand the reasons for lower sales and identify areas for improvement in both product and marketing strategy.
5. Promotions and Discounts: Run targeted promotions, especially in off-peak seasons, to increase interest and generate sales.

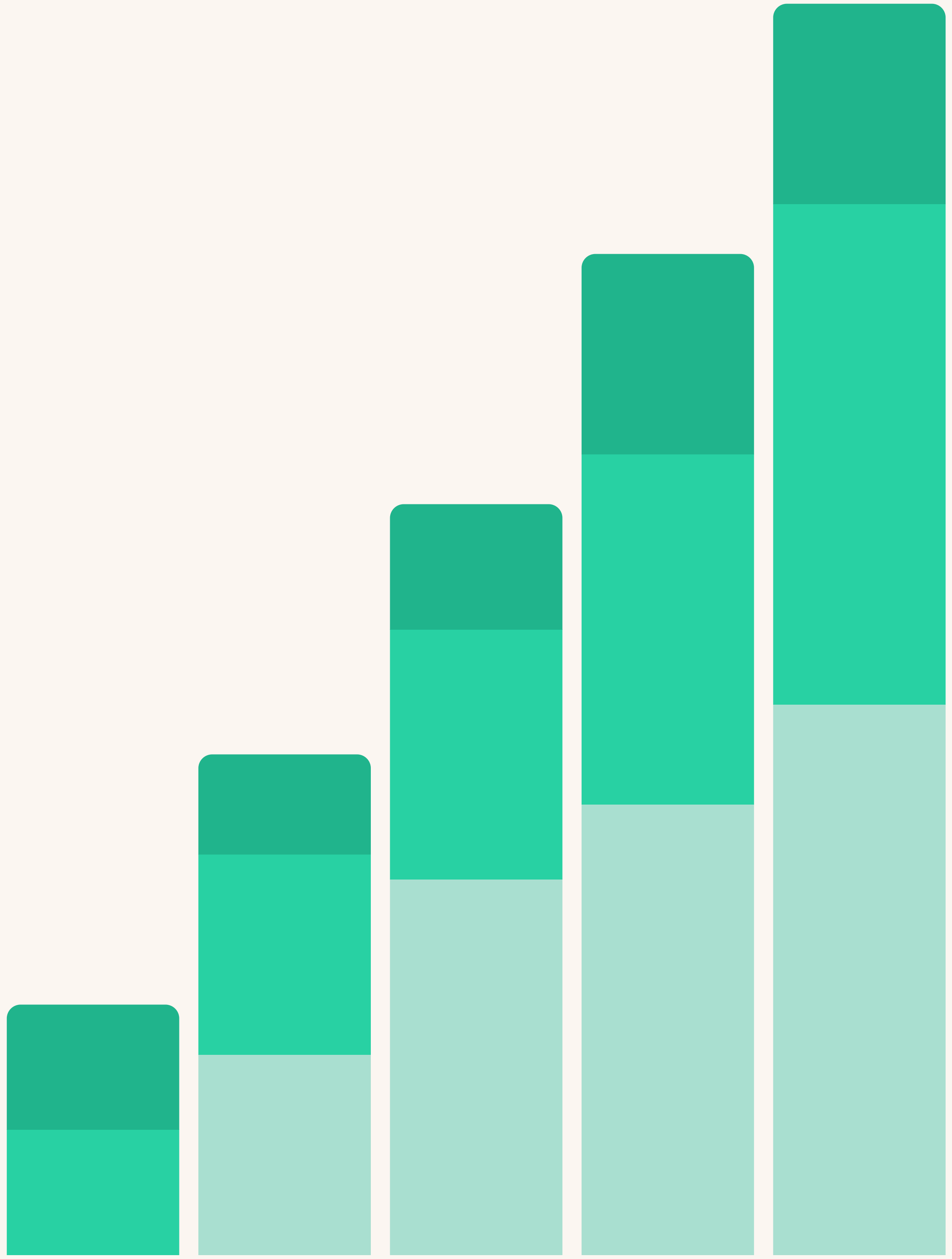
Observation

- WWI Desktop PCs generate the highest revenue, indicating strong sales and demand for high-performance computing products.
- Adventure Works has the lowest revenue, suggesting that this product category may not be resonating well with the current customer base.

Submission Details

1. **Github Repositories:**https://github.com/Aparajitha01/Plot-a-thon/blob/995bf1fc7c7b4d8b8d31999265e0efbe413c4e07/README.md
2. **Files Submitted:**

- A_Square_Visuals.pdf
- A_Square.pptx
- ReadME.md



Thank you very much!